Path dependency and cement-industry decarbonisation in the UK

The Peak Cluster CCS case — policy lock-in, abatement economics and social licence

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THE QUESTION

Can a UK industrial CCS cluster close the gap between carbon-price incentives, full-chain abatement cost and social licence — or does path-dependent policy lock cement into a high-cost pathway that still fails to earn public consent?

AT A GLANCE

INDICATOR	RESULT
Institutional lock-in (15 policy/industry documents)	**Mean 4.4 / 7 · Peak Cluster materials = 7**
Pathway-diversification or sunset clauses	**Present in 13% of the corpus**
Median LCO_2A, post-combustion MEA (N = 10,000)	**USD 131.4 / t (P5–P95: 113.8–149.7)**
UK ETS 2026 civil-penalty reference	**USD 65.2 / t**
Implied public support at 3 Mt/yr design throughput	**~USD 66 / t · ~USD 199 million / year**
Social licence (191 public-discourse records)	**51% scrutiny · 49% withholding · 0% approval**

IN BRIEF

- **Policy.** UK cluster sequencing and the Industrial Carbon Capture contract make CCS the central, contractable option. Sunset and diversification clauses are rare. The corpus does not merely prefer CCS; it narrows the space in which alternatives can still be chosen.
- **Economics.** The mature retrofit route (post-combustion MEA) sits about USD 66 / t above the 2026 UK ETS civil-penalty reference. Capture cost dominates uncertainty; cheaper pipelines do not close the gap. Efficiency, clinker substitution and alternative fuels abate at lower cost, but they do not remove process emissions on their own.
- **Licence.** Place identity, procedure and landscape — not carbon-price logic — structure public discourse. The project has not moved above withholding and conditional scrutiny. Path-dependent instruments can advance CCS while eroding legitimacy.

EVIDENCE

Design. Convergent parallel mixed-methods case study of Peak Cluster–Morecambe: inland cement and lime anchors, a ~200 km onshore CO_2 corridor, and Liverpool Bay / East Irish Sea storage. Three strands were run independently and joined in a joint display (final thesis Figure 13).

- **Documents.** WPR problem-representation analysis and Erickson-style lock-in indicators on 15 government and industry texts (NVivo 14; P01–P15). Mean lock-in 4.4 / 7; Peak Cluster consultation materials score 7.
- **Techno-economics.** Monte Carlo LCO_2A across five capture routes (seed 42); tornado sensitivity; ETS benchmarking; literature-based MAC for UK cement. Reported 2023 releases at four anchors total 2.248 Mt CO_2 (PRTR). Subsidy cases use the 3 Mt/yr ICC design throughput, not the PRTR total — a distinction often elided in sponsor materials.
- **Discourse / SLO.** Manual stance and frame coding plus VADER sentiment on 191 cleaned records. Dominant frames: place identity (49%) and procedure (34%). Ninety per cent of items express an explicit trust deficit toward operators or regulators.

Replication. Python (numpy, pandas, matplotlib) with an optional R check of the MEA median. Public code: github.com/Authenticscholar/peak-cluster-ccs-analysis. Figures on the public site are taken from the final submitted thesis PDF.

IMPLICATIONS

For governments and cluster sponsors, treat CCS as one option in a portfolio, not as the default that crowding-out is allowed to produce. For carbon-price design, an ETS level near USD 65 / t does not, on this evidence, make cement CCS privately viable. For engagement, social licence is a binding constraint, not a communications residual. Cement CCS is not shown to be unnecessary; it is shown to be premature as an exclusive, 15-year public commitment under present prices and present licence conditions.

SELECTED EXHIBITS (FINAL THESIS)

- Figure 1 — Study-area map: pipeline corridor and anchor facilities.
- Figure 2 — Policy lock-in heatmap (P01–P15).
- Figure 5 — Monte Carlo LCO_2A distributions by capture technology.
- Figure 7 — Tornado sensitivity, post-combustion MEA.
- Figure 8 — Comparative MAC curve, UK cement, with ETS references.
- Figure 13 — Joint display of convergence, dissonance and generative mechanism.

The full thesis is available on request. It is not posted publicly, pending university repository practice and possible chapter publication. This brief is a findings note for recruiters and research users, not a substitute for the examined dissertation.